Foundational IT Skills

A practical bridge from computer foundations to office automation, data analysis, cloud collaboration, generative AI and graphic design.

Course 1008 **Semester I** **Practicum** **Scheme 0:0:2:0** **Credits 1**

Official Source Declaration

> The handbook is grounded in the verified Revision 2026 SITTTR extracted source for Course 1008 — Foundational IT Skills. Official activities, practical assessment, equipment/software requirements and references are retained in this file; AI tools must be used with privacy, source checking and institutional permission.

The official syllabus source is authoritative for the course code, title, scheme, outcomes, modules, hours, activities, assessment and references. Any source-format irregularity must remain visible to the learner and examiner.

Course Dashboard

Course information

Field	Official value
Course code	1008
Course title	Foundational IT Skills
Semester	I
Course type	Practicum
Teaching scheme	0:0:2:0
Credits	1
Module hours	As shown in the official module coverage table
Lesson purpose	Handbook-scale study, practical preparation and examination revision

How to Use This Handbook

Before class

Read the module goals, identify unfamiliar terms and prepare the required file, equipment or physical space.

During practice

Follow the procedure, record observations and ask for supervision where the activity involves equipment, exercise, data or external platforms.

After practice

Compare the outcome with the checklist, correct errors and use the questions and revision cards for retrieval practice.

Malayalam support: □□□ module-□□ □□□□□ □□□□□□□□ concept □□□□□□□□□□□□□□. Practical activity □□□□□□□□□□□ safety, record, reflection □□□□□□ □□□□□□□□□□.

Objectives, Course Outcomes and CO–PO

Official Course Objectives

1. Familiarize students with basic functions and features of Computers, Operating Systems and Internet applications.
2. Enable students to prepare documents, spreadsheets and presentations.
3. Equip students with data visualization and automation skills using Pivot tables, slicers, graphs and charts.
4. Empower students to use generative AI tools for ethical content creation and guided technical learning.

Course Outcomes

Course Outcomes

CO	Outcome	Bloom level
CO1	Identify the basic functions and features of Computers, Operating Systems, and Internet applications.	Understand
CO2	Use stand-alone and cloud-based office tools to prepare documents, spreadsheets, and presentations.	Apply
CO3	Apply spreadsheet functions and Generative AI tools to construct interactive dashboards, automated reports, and professional visual presentations for interpreting complex datasets.	Apply
CO4	Apply generative AI and digital design platforms for guided learning and visual communication.	Apply

CO-PO correlation matrix

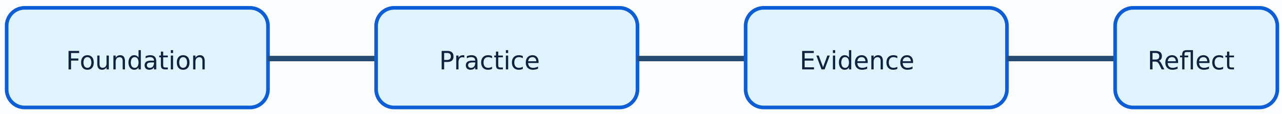
CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	3	3	-	-	-	-	-
CO3	3	3	-	3	-	-	3	-	-	-	-	-
CO4	3	3	3	3	-	-	3	-	-	-	-	-
Total	12	6	3	6	-	3	9	-	-	-	-	-
Correlation Level	3	3	3	3	-	3	3	-	-	-	-	-

Complete Syllabus Coverage

Official module and experiment coverage

Module	Official entries	Hours
I — Computer and Internet Foundations	Hardware & Software Exploration; File and Folder Management in a Linux GUI; Web & Connectivity	6
II — Office and Cloud Productivity	Professional Documentation; Spreadsheet Basics; Presentation Design; Cloud Collaboration; Series Test 1	10
III — Data and AI Presentation	Data Analysis; Data Visualization; AI Slide Generation	6
IV — Guided AI and Design	GPT Guided Learning; Graphic Design; Series Test 2; Open-Ended Experiment	8
Total	Practical entries plus tests and open-ended experiment	30

Learning Roadmap



Use the roadmap to connect individual practical tasks to course outcomes.

Practical competence grows through preparation, performance, feedback and reflection.

MODULE 1

Computer, Operating System and Internet Foundations

Official hours: 6 · CO1

Hardware, files, connectivity and responsible digital communication.

Learning goals

- Hardware and software exploration
- File and folder management in a Linux GUI
- Web and connectivity

Key facts

- Use clear file names and verify before deleting.
- Prefer reproducible formulas and recorded procedures.
- Select visuals based on the question being answered.
- Verify AI output and protect private information.

Engineering relevance: Engineering students use digital tools to store reports, data, drawings and team records.

Hardware and software exploration

Identify CPU, RAM, HDD/SSD, ports, system software and application software. Record each item and follow power/static safety.

Malayalam support: Task-□□□□ aim, procedure, result, source check □□□□□□ record □□□□□□□. Digital tools □□□□□□□□□□□□□□□ privacy□□□ ethical use□□ □□□□□□□□□.

Record: Aim · requirements · procedure · observation · result · reflection.

File and folder management in a Linux GUI

Create a semester directory tree; create, copy, move, rename and delete test files; verify paths and recoverability.

Malayalam support: Task-□□□□ aim, procedure, result, source check □□□□□□ record □□□□□□□. Digital tools □□□□□□□□□□□□□□□ privacy□□□ ethical use□□ □□□□□□□□□.

Record: Aim · requirements · procedure · observation · result · reflection.

Web and connectivity

Use advanced search operators, email etiquette, privacy controls, LMS participation and source evaluation.

Malayalam support: Task-□□□□ aim, procedure, result, source check □□□□□□□ record □□□□□□□□. Digital tools □□□□□□□□□□□□□□□□□ privacy□□□ ethical use□□ □□□□□□□□□□.

Record: Aim · requirements · procedure · observation · result · reflection.

MODULE 2

Office Automation and Cloud Collaboration

Official hours: 10 · CO2

Professional documents, formula-driven spreadsheets, presentations and version history.

Learning goals

- Professional documentation
- Spreadsheet basics
- Presentation design
- Cloud collaboration

Key facts

- Use clear file names and verify before deleting.
- Prefer reproducible formulas and recorded procedures.
- Select visuals based on the question being answered.
- Verify AI output and protect private information.

Engineering relevance: Engineering students use digital tools to store reports, data, drawings and team records.

Professional documentation

Use page layout, Heading 1/2 styles, a table of contents, captions, spelling checks and stable PDF export.

Malayalam support: Task-[illegible] aim, procedure, result, source check [illegible] record [illegible]. Digital tools [illegible] privacy[illegible] ethical use[illegible] [illegible].

Record: Aim · requirements · procedure · observation · result · reflection.

Spreadsheet basics

Use SUM, AVERAGE, COUNT, COUNTIF, IF and cell references. Replace manual repetition with reproducible formulas.

Malayalam support: Task-[illegible] aim, procedure, result, source check [illegible] record [illegible]. Digital tools [illegible] privacy[illegible] ethical use[illegible] [illegible].

Record: Aim · requirements · procedure · observation · result · reflection.

Presentation design

Create a five-slide pitch deck with one idea per slide, consistent hierarchy, restrained transitions and accessibility checks.

> **Malayalam support:** Task-□□□□ aim, procedure, result, source check □□□□□□ record □□□□□□□. Digital tools □□□□□□□□□□□□□□□ privacy□□□ ethical use□□ □□□□□□□□□.

> **Record:** Aim · requirements · procedure · observation · result · reflection.

Cloud collaboration

Use Google Docs/Sheets version history, comments, access permissions and recovery of earlier versions.

> **Malayalam support:** Task-□□□□ aim, procedure, result, source check □□□□□□ record □□□□□□□. Digital tools □□□□□□□□□□□□□□□ privacy□□□ ethical use□□ □□□□□□□□□.

> **Record:** Aim · requirements · procedure · observation · result · reflection.

MODULE 3

Data Analysis, Visualization and AI Presentation

Official hours: 6 · CO3

Pivot Tables, slicers, chart selection and carefully verified AI-assisted slide creation.

Learning goals

- Data analysis

- Data visualization
- AI slide generation

Key facts

- Use clear file names and verify before deleting.
- Prefer reproducible formulas and recorded procedures.
- Select visuals based on the question being answered.
- Verify AI output and protect private information.

Engineering relevance: Engineering students use digital tools to store reports, data, drawings and team records.

Data analysis

Clean headers, create Pivot Tables, use a Region slicer, record Grand Total, add rows and refresh.

Malayalam support: Task-□□□□ aim, procedure, result, source check □□□□□□□ record □□□□□□□□. Digital tools □□□□□□□□□□□□□□□□ privacy□□□ ethical use□□ □□□□□□□□□□.

Record: Aim · requirements · procedure · observation · result · reflection.

Data visualization

Select pie, bar, line or scatter charts according to composition, comparison, trend or relationship questions.

> **Malayalam support:** Task-□□□□ aim, procedure, result, source check □□□□□□ record □□□□□□□. Digital tools □□□□□□□□□□□□□□□ privacy□□□ ethical use□□ □□□□□□□□□.

> **Record:** Aim · requirements · procedure · observation · result · reflection.

AI slide generation

Prepare structured prompts, verify claims against source facts, edit layout and citations, and protect private data.

> **Malayalam support:** Task-□□□□ aim, procedure, result, source check □□□□□□ record □□□□□□□. Digital tools □□□□□□□□□□□□□□□ privacy□□□ ethical use□□ □□□□□□□□□.

> **Record:** Aim · requirements · procedure · observation · result · reflection.

MODULE 4

Generative AI Guided Learning and Graphic Design

Official hours: 6 · CO4

Prompt-guided technical learning and visual communication using ethical AI practices.

Learning goals

- GPT guided learning
- Graphic design

Key facts

- Use clear file names and verify before deleting.
- Prefer reproducible formulas and recorded procedures.
- Select visuals based on the question being answered.
- Verify AI output and protect private information.

Engineering relevance: Engineering students use digital tools to store reports, data, drawings and team records.

GPT guided learning

Write a Study Assistant prompt using context, role, task and output requirements; ask for uncertainty and verify the answer.

Malayalam support: Task-□□□□ aim, procedure, result, source check □□□□□□□ record □□□□□□□□. Digital tools □□□□□□□□□□□□□□□□□□ privacy□□□ ethical use□□ □□□□□□□□□□□.

Record: Aim · requirements · procedure · observation · result · reflection.

Graphic design

Create a technical event poster with hierarchy, contrast, alignment, readable text, usable images and a clear call to action.

Malayalam support: Task-□□□□ aim, procedure, result, source check □□□□□□ record □□□□□□□. Digital tools □□□□□□□□□□□□□□□ privacy□□□ ethical use□□ □□□□□□□□□.

Record: Aim · requirements · procedure · observation · result · reflection.

Formula, Procedure and Safety Bank

```
SUM(range) · AVERAGE(range) · COUNT(range) · COUNTIF(range, criteria) · IF(test, value_if_true, value_if_false)
```

Given: Runs are in B2:B21 and selection threshold is 50.
Required: Total, mean, count and selection status.
Calculation / procedure: Use `=SUM(B2:B21)`, `=AVERAGE(B2:B21)`, `=COUNT(B2:B21)`, `=COUNTIF(C2:C21,"Batsman")` and `=IF(B2>=50,"Selected","Waitlist")`.
Answer / expected result: The worksheet calculates a repeatable result.

Diagram Library for Rapid Revision

Learning roadmap

Foundation → practice → evidence → reflection.

Module diagrams

Each module contains a visual or structured checklist where it improves understanding.

See the roadmap and embedded workflow diagrams. Use the module checklists for rapid practical revision.

Official Self-learning Activities

Official self-learning activities

Week	Module	Activity	Hours
1	I	Label lab-PC ports; compare RAM, processor and storage with a smart device; list system and application software.	1
2	I	Create a semester directory tree and complete create, move, copy and delete operations.	1
3	I	Use filetype/domain search; join an LMS; draft a formal workshop email with privacy protection.	1
4	II	Create an assignment cover sheet and export professional biodata to PDF using styles.	1
5	II	Use SUM, AVERAGE, COUNTIF, IF and COUNT on the cricket dataset.	1
6	II	Edit a Slide Master so name and roll number appear on every slide.	1
7	II	Use version history to restore a deleted table and identify the editor; create an optional Overleaf document.	2
8	III	Create Pivot Tables, filter with Region slicer, record Grand Total and refresh after adding rows.	2
9	III	Create a 3D pie chart and clustered horizontal bar chart using an appropriate chart style.	2
10	III	Use Gamma.app for an eight-slide deck and Manus AI to research laptop prices and prepare a recommendation presentation.	1
11	IV	Ask ChatGPT five computer interview questions; explain C-R-T-O and write a prompt using it.	1
12	IV	Design a college art-festival logo and professional poster using Canva or a comparable platform.	1

Assessment Methodology

Practical assessment

Component	Timing	Marks
Continuous evaluation	Every lab slot	50 converted to 30
CA1 practical test	7th week; 2 hours	40 converted to 15
CA2 practical test	14th week; 2 hours	40 converted to 15
Open-ended experiment	15th week; 3 hours	50 converted to 10
ESE practical	3 hours	40
Total	—	CIE 60 + ESE 40 = 100

Malayalam support: Practical record-□ preparation, procedure, observation, result, viva, discipline, safety, teamwork □□□□□□ □□□□□□□□□□□ □□□□□□□□□□□□.

Module-wise Questions and Answers

What is a Pivot Table?

A Pivot Table is a summary view that groups structured records so categories and values can be compared quickly; it does not replace clean source data.

Why verify AI-generated slides?

AI may produce incorrect claims, labels, images or citations. Compare every important statement with the supplied source and revise before presentation.

What should a professional email contain?

A meaningful subject, greeting, concise purpose, clear request or information, professional closing and privacy-aware recipient handling.

Practice Model Paper — Not an Official Examination Paper

Instructions: Answer all short questions and attempt the practical tasks using the official record format. This model is for practice only.

1. Define one key term from Module I and explain one safety precaution.
2. Describe a complete procedure from Module II with observations and expected result.
3. Compare two methods or tools from Module III.
4. Write a short reflection showing teamwork, discipline or ethical digital practice from Module IV.
5. Design a practical record page containing aim, requirements, procedure, observation, result and reflection.

Quick Revision

Module checklist

- Hardware → operating system → files → web and email.
- Styles → formulas → presentation → collaboration.
- Pivot Tables → slicers → charts → AI slide verification.
- Prompt structure → poster hierarchy → ethics and privacy.

Common mistakes

- Deleting before verifying the file path.

- Typing repeated spreadsheet results instead of formulas.
- Using decorative charts without labels or scale checks.
- Trusting generated AI content without source verification.

Glossary

Important technical terms

Term	Short meaning
Operating system	Software that manages computer hardware and provides services for applications.
Directory tree	A hierarchical organisation of folders and files.
Pivot Table	A structured summary of records grouped by fields.
Slicer	A visible interactive filter for a data summary.
Version history	A record of edits that supports review and recovery.
Prompt	A structured instruction given to an AI system.
Generative AI	A system that produces content from an input specification.
Chart	A visual representation selected to answer a data question.

Official References and Resources

Official resources

Resource	Use
Optional document collaboration	
AI-assisted presentation activity	
AI-assisted research/presentation activity	
Graphic-design activity	

Official equipment and software

Category	Requirement
PC	Intel Core i5/AMD Ryzen 5 or higher, 16GB RAM, 512GB SSD; 30 units
Internet	Minimum 100 Mbps dedicated or broadband connection

Category	Requirement
Operating system	Windows 10/11 Professional or latest Ubuntu LTS
Office and cloud tools	Microsoft 365 or LibreOffice/OpenOffice; Google Workspace; Overleaf
AI/design tools	Access to ChatGPT/Gemini and AI-powered design platforms as available

In-page Search